

# Laser ablation absorption spectroscopy for isotopic analysis of plutonium: Spectroscopic properties and analytical performance

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## ABSTRACT

Spectroscopic properties of atomic species of plutonium were investigated by combining laser ablation and resonance absorption techniques for the analysis of a plutonium oxide sample. For 17 transitions of Pu atoms and ions, the absorbance, isotope shift, and hyperfine splitting were determined via Voigt profile fitting of the recorded absorption spectra. Three transitions were selected as candidates for analytical use. Using these transitions, we investigated the analytical performance that was attainable and determined a correlation coefficient  $R^2$  between the absorbance and plutonium concentration of 0.9999, a limit of detection of 30–130 ppm, and a relative standard deviation of approximately 6% for an abundance of  $^{240}\text{Pu}$  of 2.4%. These results demonstrate that laser ablation absorption spectroscopy is applicable to the remote isotopic analysis of highly radioactive nuclear fuels and waste materials containing multiple actinide elements.

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## 1. Introduction

Laser ablation (LA) is a phenomenon in which surface material is evaporated from a sample irradiated by intense laser pulses, so that a laser-induced plasma (LIP) develops as a result of subsequent absorption of laser energy. Because of its many attractive features, various analytical techniques employing the LA process have currently gained popularity. One typical example is a combination of LA sampling and inductively coupled plasma mass spectrometry (LA-ICP-MS) [1]. Because there is little or no need for sample preparation, this technique has been widely used for the determination of multiple elements in solid samples. For a similar reason, and owing to its simpler instrumentation, laser-induced breakdown spectroscopy (LIBS), by which the elemental compositions of samples can be determined from emissions from LIPs, has also been actively studied over the past few decades [2]. In addition to elemental analysis, some attempts have been made to employ LIBS in the analysis of isotopic ratios; in order to increase its spectral resolution and analytical precision, improvements in the measurement procedure and the selection of suitable emission lines of atomic or molecular species have been reported so far [3–9]. Despite these efforts, the number of isotopes in the determination of which LIBS is applicable is still quite limited.

Laser ablation absorption spectroscopy (LAAS), which we employed in the present study, is another LA-based technique applicable to

isotopic analysis [10–15]. In this technique, a LIP cooled by ambient gas is illuminated by a probe laser beam to excite ground-state atoms. In the cooled plasma, charged species disappear as a result of recombination with thermal electrons, and the ablated species also lose kinetic energy; the decrease in Doppler and Stark broadening leads to higher spectral resolution of LAAS. Furthermore, atoms in various excited states decay to the ground state, so that the resonance absorption of ground-state atoms is significantly enhanced. As a result, it is expected that the absorption signal of a particular isotope can be measured with a higher signal-to-background ratio, and thus the isotopic abundance can be determined precisely on the basis of proportionality between the absorbance and line-of-sight number density of the target isotopic species.

These LA-based analytical techniques have many potential applications in the field of nuclear technology because of their capability for rapid and remote (contactless) analysis of radioactive samples. In fact, there are many studies that report the employment of LIBS in the analysis of nuclear fuels and diagnosis of nuclear reactors [2,16–26]. In these applications, detailed information on isotopic compositions is frequently required. In particular, if highly radiotoxic nuclear fuels containing various actinides and fission products become widely used to increase nuclear security [27–28], the remote analysis of isotopes will be needed to safeguard verification and in process control for the fabrication of fuels. Furthermore, in particular in Japan, nuclide analysis of the molten core debris from Fukushima will be required in the near future. To treat huge amounts of radioactive waste contaminated with debris, remote analysis using an LA-based process is expected to be needed. To meet

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these needs, we have developed LIBS and LAAS for the analysis of fuel materials. Thus far, we have studied the plume dynamics of lanthanide elements as surrogates for fuel materials to find experimental conditions suitable for these analyses, e.g., types of buffer gas, gas pressure, ablation laser energy, and the height and timing of measurements [29–33]. On the basis of the results, in the present study we demonstrated the use of LAAS for the analysis of Pu isotopes.

For spectroscopic analysis such as LIBS and LAAS, knowledge of the spectroscopic properties of the analyte is of great importance. The atomic spectroscopy of Pu has been studied since the 1950s, and most of the early results have been comprehensively summarized by Blaise et al. [34–35]. Furthermore, data on the hyperfine structure (hfs) of  $^{239}\text{Pu}$  have been reported by several researchers [36–39]. In recent years, a compilation of spectroscopic data on Pu has appeared on the homepage of Laboratoire Aimé Cotton [40]. Because of these efforts, the spectroscopic properties of atomic Pu are already known to some extent. However, some data needed for the highly sensitive and isotope-selective LAAS of Pu remain unknown. For example, data on absorption strength are unavailable for most of the transitions of Pu, although relative emission intensities using various types of light source and partial absorption strengths have been reported [41]. Furthermore, hfs data remain scarce for many transitions, even those of the abundant isotope  $^{239}\text{Pu}$  [36–38].

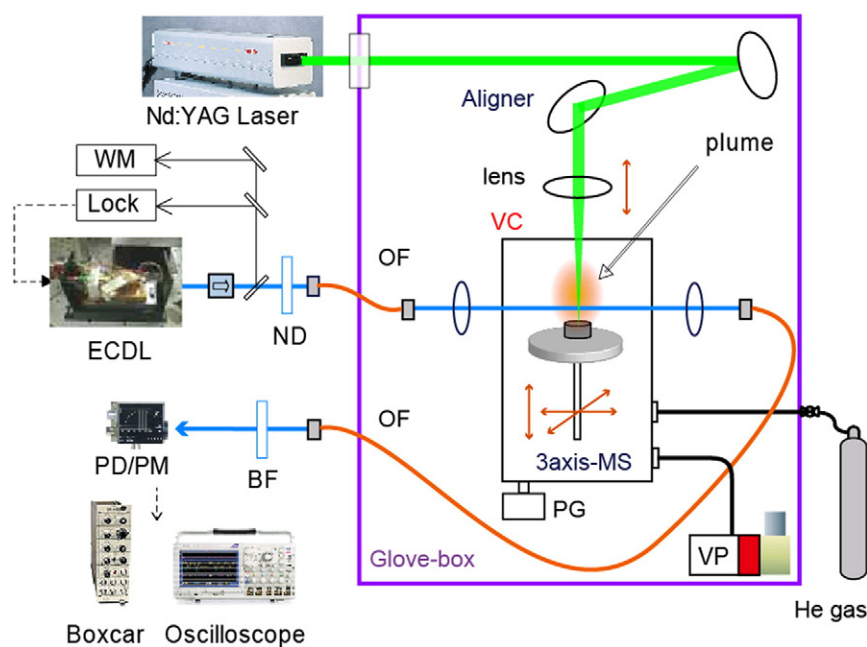
In the present study, we revealed the spectroscopic properties of some transitions from the ground states of Pu atoms and ions to select transitions suitable for LAAS. Furthermore, by using sintered uranium–plutonium mixed oxide (MOX) pellets, the elemental composition of which is close to that of actual MOX fuels, we investigated the analytical performance attainable in the LAAS of Pu. From these experiments, it is demonstrated that precise determination of isotopic abundance of Pu requires a comprehensive knowledge of spectroscopic properties such as hfs and isotope-shift.

## 2. Experimental

Fig. 1 shows the experimental setup. LA was accomplished using a Q-switched neodymium:yttrium aluminum garnet (Nd:YAG) laser (Quantel Brilliant EaZy, pulse duration: 5 ns, repetition rate: 10 Hz).

The second harmonics (532 nm, 0.15 mJ/pulse) of the laser was directed into a large glove box (size:  $1 \times 3 \times 2$  m) used for treating nuclear fuel materials in the NUCF-BECKY facility at the JAEA Tokai site. Inside the glove box, the path of the laser beam was automatically aligned using a beam alignment system (TEM Messtechnik Aligna). The beam, which passed through a biconvex lens ( $f = 200$  mm), was introduced into a vacuum chamber installed in the glove box to irradiate a target sample at normal incidence. The lens holder was mounted on a linear translation stage (Thorlabs MT1-Z8), and the lens–sample distance was adjusted so as to fix the focal point at a position approximately 1 mm behind the sample surface. The diameter of the beam on the surface was approximately 200  $\mu\text{m}$ . The sample was mounted on a three-axis motion control stage (Tsuji Electronics PM16C-04XDL) to avoid the crater effect.

Prior to the measurements, the chamber was evacuated, flushed with helium gas several times, and then filled with helium gas (purity: 99.99%) at a pressure of 800 Pa. The pressure was monitored using a convection-enhanced Pirani gauge (Granville-Phillips 475series). The ablation plume produced from the sample was intersected by a probe laser beam generated from a continuous-wave external-cavity diode laser (ECDL). The beam, which was introduced into the glove box through an optical fiber (core 200  $\mu\text{m}$ , 10 m long), was collimated to a diameter of approximately 800  $\mu\text{m}$  by a biconvex lens ( $f = 75$  mm) and propagated at a height of 2–2.5 mm above the sample surface. The ECDL head was replaced depending on the desired wavelength range. Table 1 summarizes the specifications of all the laser diodes that were utilized. The power of the illuminating laser was set at 0.05–0.1 mW by a variable neutral-density filter to ensure clear absorption. The laser frequency was controlled using the fringe offset-locking technique [42] with a scanning Fabry–Pérot interferometer (free-spectral range: 300 MHz; Finesse: 200) and a frequency-stabilized He–Ne laser (REO model 32,734). Frequency stabilization and scanning were performed by varying the voltage applied to the piezoactuator to adjust the length of the external cavity. The typical instantaneous bandwidth and residual frequency fluctuation under locking conditions were estimated to be approximately 1 MHz and several MHz/h, respectively, both of which were sufficiently less than the observed absorption



**Fig. 1.** Schematic illustration of experimental setup for Pu-LAAS. VP, PG, BF, PD, PM, ND, WM and 3axis-MS denote vacuum pump, convection-enhanced Pirani gauge, band-pass filter, fast photodiode, photomultiplier, neutral density filter, wavemeter, and three axis motion control stage, respectively. Vacuum chamber is installed inside a large glove-box for treating nuclear fuel materials.

**Table 1**  
Specifications of laser diodes used in this study.

| Manufacturers | Model number | $\lambda$ (nm) | P(mW) |
|---------------|--------------|----------------|-------|
| Nichia        | NDU1166      | 375.6          | 20    |
| Nichia        | NDUA116T     | 395.8          | 20    |
| Nichia        | NDV4212      | 405.3          | 100   |
| Nichia        | NDHV310ACAE1 | 413.0          | 30    |
| Roithner      | RLT420-50CMG | 419.5          | 50    |
| Nichia        | NDB4116      | 440.0          | 100   |
| Osram         | PL450B       | 449.0          | 80    |
| Opnext        | HL6385DG     | 641.5          | 150   |
| Opnext        | HL6501MG     | 656.9          | 35    |
| Toshiba       | TOLD9215     | 673.7          | 10    |
| Mitsubishi    | ML101J08     | 662.0          | 40    |
| Mitsubishi    | ML101J10-01  | 688.0          | 50    |

The symbols  $\lambda$  and P denote center wavelength and maximum output power, respectively.

linewidth (>350 MHz). The absolute wavelength and longitudinal mode of the laser were continuously monitored using a Fizeau wavemeter (HighFinesse WSU-10).

The probe beam, after passing through the plume, was directed out of the glove box through another fiber (core 200  $\mu$ m, 10 m long). The intensity of the beam, which was passed through an interference filter (bandwidth 10 nm) to block plasma emissions, was measured with a fast photoreceiver (Femto HCA-S-200 M) or photomultiplier (Hamamatsu R1104). When the probe laser was tuned to the resonant wavelength of Pu, the transmitted intensity decreased temporarily as the plume intersected the probe beam. The temporal transmittance profile was averaged and recorded with a digital oscilloscope. Furthermore, the absorption spectrum was recorded by scanning the wavelength of the probe laser (1.2 GHz/min) and recording the signal transmitted in a particular time interval with a boxcar integrator (Stanford Research System SR250) with an integration time of 1 s (i.e., 10 laser shots). The spectral resolution achieved by this signal processing technique was estimated to be approximately 20 MHz, which was much less than the observed absorption linewidth (>350 MHz) of the ablated species.

In the present study, U-Pu MOX samples were used as surrogates for an actual MOX fuel. Table 2 lists the elemental compositions of the samples. The isotopic abundance of Pu is approximately 97.6 wt%  $^{239}\text{Pu}$ , 2.4 wt%  $^{240}\text{Pu}$ , and negligible amounts of other isotopes. After crushing and mixing  $\text{PuO}_2$  and  $\text{UO}_2$  powders, pellets (diameter 6 mm, thickness: 1–2 mm) were prepared by compressing the powders under a hydrostatic pressure of approximately 16 MPa. These pellets were subsequently sintered at 1400  $^\circ\text{C}$  for 16 h under an atmosphere of 4%  $\text{H}_2$  mixed with He gas using a siliconite furnace in the TRU-HITEC facility at JAEA.

Table 3 summarizes the previously reported data for the transitions investigated in this study, i.e., the resonance wavelength in air, energy and total angular momentum ( $J$ -value) of the relevant upper level, relative emission intensity, hfs splitting of the upper level of  $^{239}\text{Pu}$ , and transition isotope shift (IS) ( $^{240}\text{Pu}$ – $^{239}\text{Pu}$ ). Transitions involving the ground states of atoms and ions are reported to have relatively higher emission intensities for electrodeless discharge lamps. Because the lowest metastable states of Pu atoms and ions lie in the energy range higher than 2000  $\text{cm}^{-1}$ , their population in the ablation plume is expected to be low at a delay time of 5–10  $\mu\text{s}$  [29]. (As will

**Table 2**  
Elemental composition in U-Pu mixed oxide samples used in this study.

| Sample  | Total weight [mg] | Pu [wt%] | U [wt%] | O [wt%] |
|---------|-------------------|----------|---------|---------|
| 88%-Pu  | 248               | 88.2     | 0.0     | 11.8    |
| 27%-Pu  | 263               | 26.5     | 61.7    | 11.8    |
| 9%-Pu   | 261               | 8.9      | 79.3    | 11.8    |
| 0.4%-Pu | 267               | 0.4      | 87.8    | 11.8    |

be discussed in Section 3.1.3, the measured gas-kinetic temperature is only 360 K at a delay time of 5–10  $\mu\text{s}$ ). Thus, we only measured the transitions from the ground states in the present study. Moreover, the studied wavelength range was limited to the range in which commercially available laser diodes can emit without the use of a harmonic generation technique. Because the  $J$ -values for the ground states of atoms and ions are 0 and 1/2, respectively, the  $J$ -values of the upper levels accessible from the ground states are 1 for atoms and 1/2 or 3/2 for ions, as listed in Table 3.

### 3. Results and discussion

#### 3.1. Absorption spectra and hyperfine structure

##### 3.1.1. Temporal profile of transmittance

Fig. 2 shows typical temporal profiles of transmittance signals recorded for Pu atoms and ions. To avoid absorption saturation, some transitions were measured with a lower pulse energy and/or lower gas pressure, whereas most transitions were measured under identical conditions such as a pulse energy of 0.15 mJ and a helium gas pressure of 800 Pa. From the measured transmittance ( $I/I_0$ ), the absorbance ( $Abs$ ) was obtained using the following Lambert–Beer law:

$$Abs = -\log_{10} \left( \frac{I}{I_0} \right) \quad (1)$$

where  $I_0$  and  $I$  denote the intensities of the incident and transmitted beams, respectively. Apparently, the absorbance of ions decreases more rapidly than that of neutral atoms. This difference arises from the recombination and rapid expansion of ionic species, as reported previously for other elements [30]. Thus, during the recording of the spectrum, the time gate of the boxcar integrator (gate width: 1  $\mu\text{s}$ ) was delayed by 5  $\mu\text{s}$  from the onset of the ablation pulse for the signal of ionic Pu and by 10  $\mu\text{s}$  for the signal of neutral Pu. This delay is also of great importance for avoiding the transient effect, which will be discussed later.

##### 3.1.2. Typical spectra of ablated Pu ions

Because the electronic angular momentum of the ground state of Pu ions is  $J_1 = 1/2$ , there are two allowed transitions to the upper states, for which  $J_2 = 1/2$  and  $3/2$ . As an example of the former case, Fig. 3(a) shows the absorption spectrum recorded at around 453.6 nm. Owing to its intense emission from an ICP source, Edelson et al. used this transition to determine the limit of detection (LOD) of the ICP–atomic emission spectroscopy of Pu (see Figs. 4 and 6 in [39]). In the case of  $^{239}\text{Pu}$ , which has non-zero nuclear spin ( $I = 1/2$ ), the ground state, as well as the upper state, splits into two hfs levels, for which  $F_2 = 0$  or 1, where  $F$  is quantum number for total angular momentum. Because the ( $F_1 \rightarrow F_2$ ) = (0  $\rightarrow$  0) transition is forbidden, this transition splits into three lines, namely, (0  $\rightarrow$  1), (1  $\rightarrow$  1), and (1  $\rightarrow$  0). These splitting intervals are normally much smaller than the fine structure splitting, and thus they are calculated in the first order perturbation theory (Casimir's formula). This leads to:

$$W_F = W_J + \frac{1}{2}CA + \frac{3C(C+1)-4I(I+1)J(J+1)}{8I(2I-1)J(2J-1)}B \quad (2)$$

where

$$C = F(F+1)-I(I+1)-J(J+1) \quad (3)$$

$W_F$  and  $W_J$  denote the energies of individual hfs levels and the energy of the center of gravity (COG) of all hfs levels, and  $A$  and  $B$  denote the electronic–nuclear magnetic dipole and electric quadrupole coupling

**Table 3**

Relative absorbance, hyperfine splitting, and isotope shift for transition from ground states of Pu.

| $\lambda_{\text{air}}(\text{nm})^{\text{a)}$ | upper( $\text{cm}^{-1}$ ) <sup>b)</sup> | $J_2^{\text{a)}$ | Present study |                           |                           |             |                   | Literature                              |                              |
|----------------------------------------------|-----------------------------------------|------------------|---------------|---------------------------|---------------------------|-------------|-------------------|-----------------------------------------|------------------------------|
|                                              |                                         |                  | Abs           | HFS <sub>U</sub><br>(MHz) | HFS <sub>L</sub><br>(MHz) | IS<br>(MHz) | Int <sup>a)</sup> | HFS <sub>U</sub> <sup>b)</sup><br>(MHz) | IS <sup>b),c)</sup><br>(MHz) |
| <i>(a) Pu atom</i>                           |                                         |                  |               |                           |                           |             |                   |                                         |                              |
| 375.3628                                     | 26,633.288                              | 1                | 23*           | 1000                      | –                         | – 3620      | 10,000            |                                         | – 3660                       |
| 393.6320                                     | 25,397.208                              | 1                | 0.40          | ~0                        | –                         | x           | 300               |                                         | – 360                        |
| 397.9460                                     | 25,121.896                              | 1                | 0.43          | 1040                      | –                         | x           | 300               |                                         | – 1110                       |
| 403.9024                                     | 24,751.439                              | 1                | 1.3*          | – 800                     | –                         | x           | 300               |                                         | 300                          |
| 405.5095                                     | 24,653.345                              | 1                | 1.2*          | ~0                        | –                         | x           | 100               |                                         | 180                          |
| 413.3005                                     | 24,188.639                              | 1                | 0.18          | 620                       | –                         | – 1820      | 10,000            |                                         | – 1920                       |
| 420.6481                                     | 23,766.139                              | 1                | 2.2*          | 3870                      | –                         | – 4550      | 10,000            | 3840                                    | – 4650                       |
| 440.3047                                     | 22,705.158                              | 1                | 0.16          | – 1050                    | –                         | x           | 300               |                                         | – 30                         |
| 440.9510                                     | 22,671.890                              | 1                | 0.10          | ~0                        | –                         | 3180        | 100               |                                         | 3210                         |
| 648.8853                                     | 15,406.760                              | 1                | 1.8*          | 1270                      | –                         | 3680        | 10,000            | 1275                                    | 3720                         |
| 677.1497                                     | 14,763.705                              | 1                | 0.03          | 540                       | –                         | x           | 1000              |                                         | 5850                         |
| <i>(b) Pu ion</i>                            |                                         |                  |               |                           |                           |             |                   |                                         |                              |
| 398.8502                                     | 25,064.975                              | 3/2              | 0.13          | 930                       | 7480                      | x           | 1000              | 810                                     | – 2730                       |
| 412.9923                                     | 24,206.710                              | 3/2              | 0.09          | ~0                        | 7480                      | x           | 10,000            | 0                                       | – 1110                       |
| 453.6146                                     | 22,038.970                              | 1/2              | 1.2*          | 1490                      | 7410                      | x           | 10,000            | 1380                                    | – 2820                       |
| 656.1689                                     | 15,235.780                              | 1/2              | 0.01          | 1420                      | 7420                      | x           | 100               | 1680                                    | 5310                         |
| 662.1215                                     | 15,098.830                              | 3/2              | 0.04          | 2320                      | 7360                      | 12,370      | 1000              |                                         | 12,390                       |
| 686.5482                                     | 14,561.625                              | 3/2              | 0.09          | 4080                      | 7430                      | 7930        | 3000              |                                         | 7950                         |

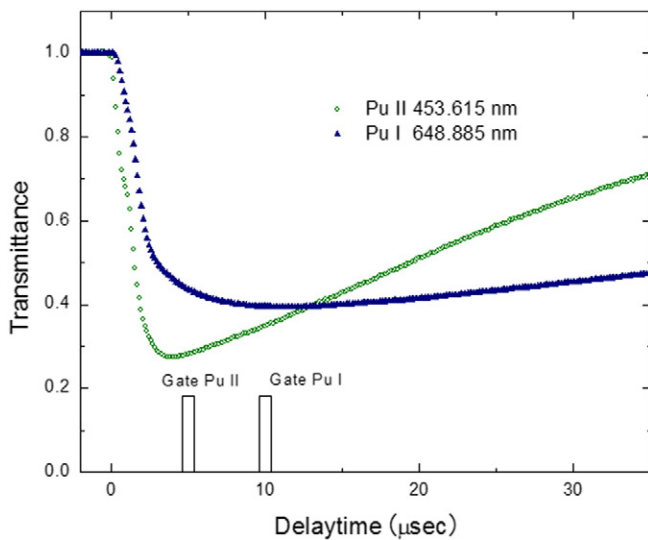
a) from [34], b) from [40], c) from [35].

Intensity (Int) in literature column denotes relative emission strength from electrode-less discharge lamp. IS denotes transition isotope-shift [ $^{240}\text{Pu}$ – $^{239}\text{Pu}$ ], and HFS<sub>U</sub> and HFS<sub>L</sub> denote hyperfine splitting of upper and lower levels, respectively. Abs means the absorbance measured for the most intense hfs component of  $^{239}\text{Pu}$ , and the values with asterisk were calculated considering the difference in the experimental conditions introduced to avoid saturating the absorption signal. The symbol x means that the IS cannot be measured because of the overlap of  $^{239}\text{Pu}$  and  $^{240}\text{Pu}$  peaks.

constants (hfs constants), respectively. Further, if we consider electronic dipole transition between two hfs levels, the line strength  $S(F_1 \rightarrow F_2)$  can be determined using the following formulae [43]:

$$S(F_1 \rightarrow F_2) \propto (2F_1 + 1)(2F_2 + 1) \left\{ \begin{matrix} J_2 & F_2 & I \\ F_1 & J_1 & 1 \end{matrix} \right\}^2 \quad (4)$$

where the symbol in the curly brackets denotes a Wigner 6J symbol. Once correct splitting schemes for the upper and lower levels are assigned, the position of the COG and the hfs constants can be determined using Eqs. (2) and (3).



**Fig. 2.** Temporal profiles of transmittance signal for ground state atoms (blue filled triangles) and ions (green open circles) of Pu along with the gate signals of boxcar integrator for spectrum measurement.

In the case of the transition at 453.6 nm, the level splitting of  $^{239}\text{Pu}$  can be expressed as follows:

$$W_{F_1=1} = W_{J_1=\frac{1}{2}} + \frac{1}{4}A \text{ and } W_{F_1=0} = W_{J_1=\frac{1}{2}} - \frac{3}{4}A \text{ for the lower level, and } W'_{F_2=1} = W'_{J_2=\frac{1}{2}} + \frac{1}{4}A' \text{ and } W'_{F_2=0} = W'_{J_2=\frac{1}{2}} - \frac{3}{4}A' \text{ for the upper level.} \quad (5)$$

Moreover, from Eq. (4) the relative intensities of each hfs peak are expected to be in the ratio of  $(1 \rightarrow 0):(1 \rightarrow 1):(0 \rightarrow 1) = 1:2:1$ .

In the actual spectrum shown in Fig. 3(a) three peaks appear, and the interval between the two right-hand peaks is nearly the same as the splitting reported for the ground state of the ions (7.47 GHz [38]). Fig. 3(c) shows the scheme of level splitting predicted from these intervals. No other candidate scheme can reproduce these observed splitting intervals. Moreover, the measured relative intensity of each hfs peak is in general agreement with that expected from Eq. (4). Therefore, we performed multiple Voigt profile function (OriginLab Co., Origin v8.6, Levenberg-Marquardt algorithm):

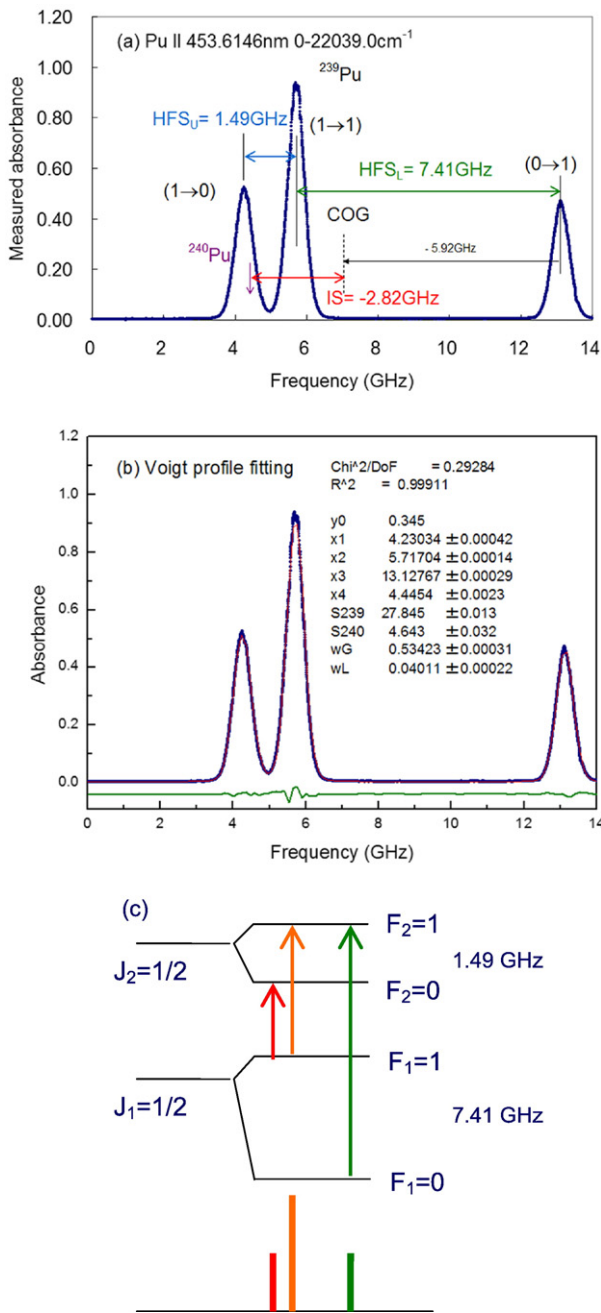
$$V(x) = V_0 + \sum_{i=1}^4 S_i \frac{2}{\pi^{3/2}} \frac{\ln 2}{w_G^2} \int_{-\infty}^{\infty} \frac{e^{-t^2}}{\left( \sqrt{\ln 2} \frac{w_L}{w_G} \right)^2 + \left( \sqrt{4 \ln 2} \frac{x - x_i}{w_G} - t \right)^2} dt \quad (6)$$

with the following constraint conditions:

$$S_1 = \frac{1}{2} S_2 = S_3, \quad ^{239}\text{S} = \sum_{i=1}^3 S_i, \text{ and } ^{240}\text{S} = S_4 \quad (7)$$

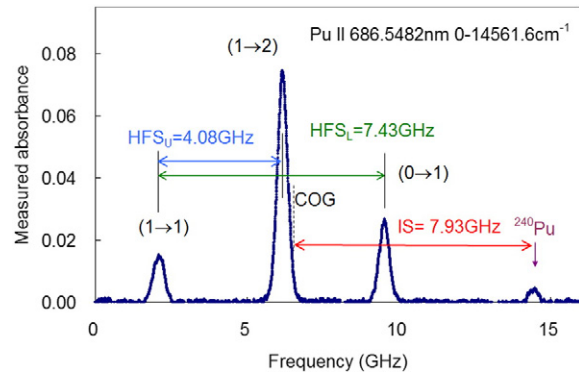
where  $V_0$  is the constant background;  $S_1$ – $S_4$  are the areas of the  $(1 \rightarrow 0)$ ,  $(1 \rightarrow 1)$ , and  $(0 \rightarrow 1)$  peaks of  $^{239}\text{Pu}$  and the peak of  $^{240}\text{Pu}$ ;  $^{240}\text{S}$  and  $^{239}\text{S}$  are the total areas of the  $^{240}\text{Pu}$  and  $^{239}\text{Pu}$  peaks, which will be used for isotopic analysis, as discussed later;  $w_G$  and  $w_L$  are the Gaussian and Lorentzian widths; and  $x_1$ – $x_4$  are the positions of the  $(1 \rightarrow 0)$ ,  $(1 \rightarrow 1)$ , and  $(0 \rightarrow 1)$  peaks of  $^{239}\text{Pu}$  and the peak of  $^{240}\text{Pu}$ , respectively.





**Fig. 3.** Absorption spectra of ground state of Pu ions around 453.6 nm. (a) Measured spectrum. The  $^{240}\text{Pu}$  peak is obscured by overlapped hfs peaks of  $^{239}\text{Pu}$ . (b) Comparison of measured (blue solid curve) and fitted (red broken curve) spectra. The lower trace (green solid curve) shows the deviation between the measured and fitted spectra. Good agreement was obtained with the fit parameters listed in this figure. (c) Expected hfs scheme reproducing the observed peak intervals and relative intensities. The expected relative intensity of this transition is  $(1 \rightarrow 0):(1 \rightarrow 1):(0 \rightarrow 1) = 1:2:1$ .

Because the  $^{240}\text{Pu}$  peak is most likely to be obscured by the overlapping  $^{239}\text{Pu}$  peaks in Fig. 3(a), the position of the  $^{240}\text{Pu}$  peak was assumed by considering the reported value of the IS. Fig. 3(b) shows the spectrum fitted according to Eq. (6) (red broken curve), together with the recorded spectrum (blue solid curve). In addition, the lower trace (green solid curve) shows the residual of the fitting. From this figure, the splitting intervals of the upper ( $\text{HFS}_U$ ) and lower ( $\text{HFS}_L$ ) levels were estimated to be 1.49 GHz and 7.41 GHz, respectively, which are listed in Table 3. Furthermore, the positions of the



**Fig. 4.** Measured absorption spectrum of ground state of Pu ion around 686.5 nm. The expected relative intensity of this transition is  $(1 \rightarrow 1):(1 \rightarrow 2):(0 \rightarrow 1) = 1:5:2$ . The rightmost peak is attributed to the peak of  $^{240}\text{Pu}$ .

COG and the  $^{240}\text{Pu}$  peak expected from the reported values of the IS [35,40] are shown in Fig. 3(a) as a vertical broken line and a downward arrow, respectively.

As was described above, from the ground state of Pu ions the transition to the upper level for which  $J_2 = 3/2$  is also allowed. In this case, the transition also splits into three lines, of which the relative intensities are expected to be in the ratio of  $(1 \rightarrow 1):(1 \rightarrow 2):(0 \rightarrow 1) = 1:5:2$ . The splitting of the upper level and the constraint conditions are expressed as follows:

$$W'_{F_2=2} = W'_{J_2=3/2} + \frac{3}{4}A', \text{ and } W'_{F_2=1} = W'_{J_2=3/2} - \frac{5}{4}A', S_1 = \frac{1}{5}S_2 \\ = \frac{1}{2}S_3, {}^{239}\text{S} = \sum_{i=1}^3 S_i, \text{ and } {}^{240}\text{S} = S_4 \quad (8)$$

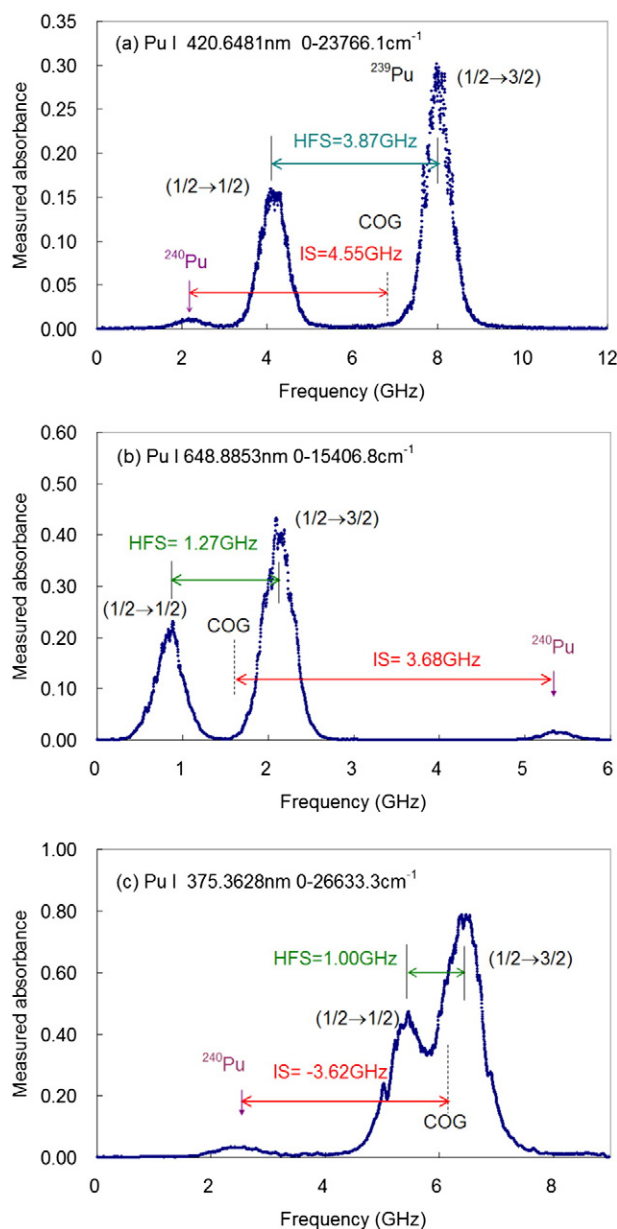
As an example of this case, Fig. 4 shows the absorption spectrum at around 686.5 nm. From curve fitting, the splitting interval of the upper level was determined to be 4.08 GHz. In addition, the transition IS was determined to be 7.93 GHz, which is in good agreement with the literature value listed in Table 3.

### 3.1.3. Typical spectra of ablated Pu atoms

The absorption spectrum of the ground state of Pu atoms is not complicated in comparison to that of ions. Fig. 5(a) shows the spectrum of the transition at 420.6 nm. Because of its higher absorbance, this spectrum was recorded with a reduced ablation energy (0.05 mJ), as well as using a sample with a lower concentration (27% Pu). According to the selection rule for an electronic dipole transition, only the transition to the upper level, for which  $J_2 = 1$ , is allowed from the ground state, for which  $J_1 = 0$ . In the case of  $^{239}\text{Pu}$ , the transition from the ground state splits into two lines, namely,  $(1/2 \rightarrow 1/2)$  and  $(1/2 \rightarrow 3/2)$ , the interval of which corresponds to the hyperfine splitting of the upper level. According to Eqs. (2) and (3), the splitting of the upper level is expressed as follows:

$$W_{F_2=3/2} = W_{J_2=1} + \frac{1}{2}A, \text{ and } W_{F_2=1/2} = W_{J_2=1} - A \quad (9)$$

The COG is located at the position that divides the interval between the  $(1/2 \rightarrow 1/2)$  and  $(1/2 \rightarrow 3/2)$  peaks in the ratio of 2:1, and the relative intensities are expected to be in the ratio of  $(1/2 \rightarrow 1/2):(1/2 \rightarrow 3/2) = 1:2$ . Considering the isotopic abundance of the Pu that was utilized, the two prominent peaks in Fig. 5(a) are attributed to  $^{239}\text{Pu}$  and the left-hand weak peak is attributed to  $^{240}\text{Pu}$ . Therefore, we performed least-squares fitting on the spectrum using a multiple Voigt function similar



**Fig. 5.** Measured absorption spectra of ground state of Pu atoms around (a) 420.6 nm, (b) 648.9 nm, and (c) 375.4 nm. Spectra (a) and (b) were measured with a sample containing 27% Pu, whereas spectrum (c) was measured with a sample containing 9% Pu. To avoid saturation of the absorption signal, these spectra were measured under experimental conditions slightly different from the standardized ones (see in text).

to Eq. (6) but replacing the term number by  $i = 3$  and the constraint condition by:

$$S_1 = \frac{1}{2} S_2, {}^{239}S = \sum_{i=1}^2 S_i, \text{ and } {}^{240}S = S_3 \quad (10)$$

where  $S_1$ – $S_3$  are the areas of the  $(1/2 \rightarrow 1/2)$  and  $(1/2 \rightarrow 3/2)$  peaks of  ${}^{239}\text{Pu}$  and the peak of  ${}^{240}\text{Pu}$ , respectively.

From this curve fitting, the hyperfine splitting of the upper level and the transition IS were determined to be 3.87 GHz ( $A = 2.58$  GHz) and  $-4.55$  GHz, respectively, which are in reasonable agreement with the literature values [35,40]. This transition exhibits a hyperfine splitting of which the magnitude is close to the IS, which results in a slight overlap of the  ${}^{239}\text{Pu}$   $(1/2 \rightarrow 1/2)$  and  ${}^{240}\text{Pu}$  peaks in the wings of the spectrum.

Fig. 5(b) shows the spectrum of the transition at 648.9 nm. Because the absorbance of this transition is almost the same as that of the transition at 420.6 nm, this spectrum was recorded under experimental conditions identical to those for Fig. 5(a). The hyperfine splitting of this transition is smaller than that in Fig. 5(a) and therefore the interval between the  ${}^{239}\text{Pu}$   $(1/2 \rightarrow 3/2)$  and the  ${}^{240}\text{Pu}$  peaks is larger, which is suitable for isotopic analysis. The influence of this interval on analytical performance will be discussed later in more detail.

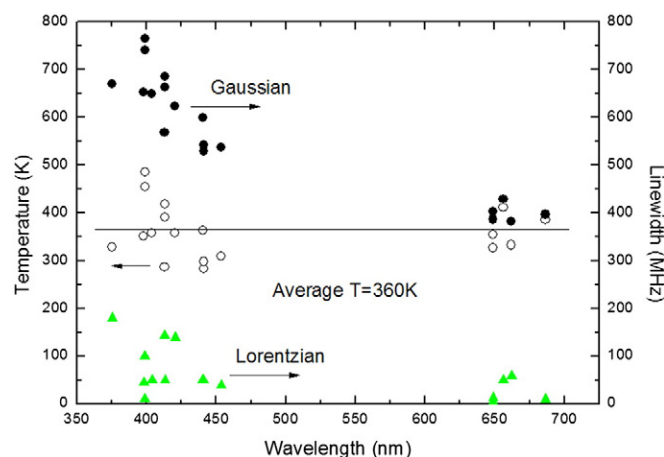
A larger interval is also found for the transition at 375.5 nm (Fig. 5(c)). Because the absorbance of this transition is the strongest of the transitions measured in the present study, and the peak absorption approaches 100% even when using a sample with a lower concentration (9% Pu), this spectrum was recorded with a lower ablation pulse energy and a longer delay time to decrease the peak absorption to approximately 80%.

It is noteworthy that no spectral interference or background bias appears in the spectra in Figs. 5(a)–(c), although the sample contains abundant U. This high elemental selectivity is one of the great advantages of LAAS in comparison to LIBS.

Table 3 summarizes the spectroscopic properties determined from the curve fitting procedure for each transition, i.e., the absorbance, hyperfine splitting of the upper and lower levels, and transition IS ( ${}^{240}\text{Pu}$ – ${}^{239}\text{Pu}$ ). The absorbance value corresponds to the most intense hfs component of  ${}^{239}\text{Pu}$  measured under identical experimental conditions (ablation energy: 0.15 mJ, He gas pressure: 800 Pa, and delay time: 5  $\mu\text{s}$  for ions and 10  $\mu\text{s}$  for atoms). In this table, some values, which are marked by asterisks, were calculated from the measured absorbance by considering the difference in the experimental conditions. Because the values of absorbance vary depending on various conditions, the given values are for guidance only.

Using the fitting parameter of the Gaussian width  $w_G$ , the dependence of the wavelength on the linewidth is shown in Fig. 6 (filled circles). From this figure, it is seen that the linewidths of the measured transitions vary in the range of 350–800 MHz, and the width increases with a decrease in wavelength. This variation most probably arises from the dependence on wavelength of Doppler broadening as follows [44]:

$$\Delta\nu = \sqrt{\frac{8kT \ln 2}{mc^2}} \times \nu \quad (11)$$



**Fig. 6.** Measured Gaussian widths (filled circles), Lorentzian widths (green filled triangles) and translational temperatures (open circles) determined on the assumption of Doppler broadening. The observed wavelength dependence of Gaussian width can be attributed to that of Doppler broadening. Solid line represents the mean of the determined temperatures.

where  $k$ ,  $T$ ,  $m$ ,  $c$ , and  $\nu$  denote the Boltzmann constant, translational temperature, atomic mass of Pu, speed of light, and frequency of atomic resonance, respectively.

The translational temperatures determined from Doppler broadening are also shown in Fig. 6. Clearly, the resulting temperature is independent of the wavelength and has an extremely low value (approximately 360 K). Because the temperature of the plasma immediately after ablation is frequently reported to be higher than 10,000 K [44–45], this result suggests that the plume is rapidly cooled via multiple collisions with gas molecules, so that most atoms and ions are in the ground states at a delay time of 5–10  $\mu$ s. Furthermore, this result indicates that the uncertainty in our determination of frequency depends on the wavelength and is approximately  $\pm 25$  MHz at around 650 nm and  $\pm 60$  MHz at around 400 nm. The values of IS listed in Table 3 are in plausible agreement with the reported values of IS within the range of these uncertainties. The hyperfine splitting of the ground state of Pu ions was determined to be  $7430 \pm 50$  MHz ( $1\sigma$ ) from the average of six sets of transition data, whereas the literature value is  $7470 \pm 60$  MHz [38].

In addition to Gaussian width, Fig. 6 shows the fitting parameters of Lorentzian width  $w_L$  as well (filled triangles). Because the Lorentzian component originated from the collisional broadening is expected to be small for the present gas condition, the magnitude is mostly determined from the natural linewidth of each transition.

Because of their stronger absorbance and larger IS, we selected three atomic transitions (375.4 nm, 420.6 nm, and 648.9 nm) as the most suitable absorption lines for the isotopic analysis of Pu. Among these, we mainly used the transition at 420.6 nm to evaluate the analytical performance of the LAAS of Pu, because this transition has been successfully used for the isotopic analysis of Pu based on other spectroscopic techniques [46–48].

### 3.2. Evaluation of analytical performance

#### 3.2.1. Linearity of calibration curve

Using the prepared MOX samples (Table 2), we constructed a calibration curve between the absorbance and Pu concentration. In Fig. 7 the absorbance was determined from the signal of the ( $1/2 \rightarrow 3/2$ ) line of the transition at 420.6 nm by averaging the signal at a delay time of 10  $\mu$ s over 200 laser shots. Throughout the present evaluation of analytical performance the absorbance was determined in this manner unless otherwise described. From least-squares fitting of this curve, the correlation coefficient was found to be  $R^2 = 0.9999$  in the concentration range below 27%. In the same way, high linearity was also ensured for the absorption lines of  $^{239}\text{Pu}$  ( $1/2 \rightarrow 1/2$ ) and  $^{240}\text{Pu}$ .

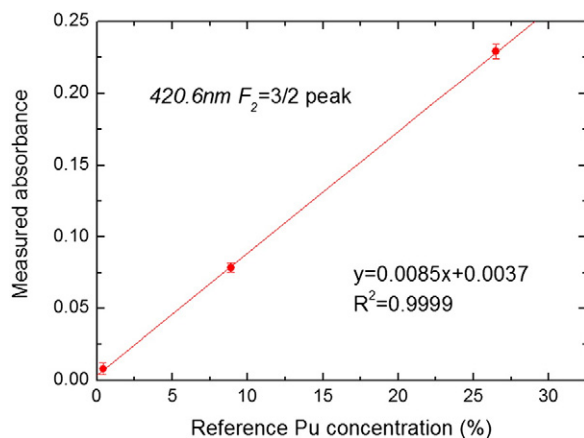


Fig. 7. Calibration curve between absorbance and Pu concentration obtained from  $^{239}\text{Pu}$  ( $1/2 \rightarrow 3/2$ ) peak of 420.6-nm transition. The number of replicate measurement is ten for each sample with different Pu concentration.

#### 3.2.2. Determination of isotopic abundance

On the basis of the linearity of the absorbance signal, the isotopic abundance of  $^{240}\text{Pu}$  was measured in two different ways: i) the use of fitting results for the spectra recorded in Sections 3.1.2 and 3.1.3 and ii) measurement of the absorption profile for the transitions of each isotope. In the former method the isotopic abundance was determined from the fitting parameters  $^{240}S$  and  $^{239}S$  for the seven transitions listed in Table 4. The abundance of  $^{240}\text{Pu}$  was calculated from the ratio of  $\frac{^{240}S}{^{239}S + ^{240}S}$ . The mean abundance of  $^{240}\text{Pu}$  for these transitions was determined to be  $2.6 \pm 0.3\%$  (relative standard deviation [RSD] 11.5%,  $1\sigma$ ).

In contrast, in the latter method we measured the temporal profile of each absorption peak (Fig. 8). Because this method requires no wavelength scan, the measurement procedure can be simplified to shorten the analysis time. However, some additional precautions are needed to obtain precise analytical results. At first, we investigated how the delay time affects the determination of isotopic abundance. In Fig. 8, the green open circles represent the absorption profile of the  $^{239}\text{Pu}$  ( $1/2 \rightarrow 3/2$ ) line of the transition at 420.6 nm, which was recorded using a sample containing 0.4% Pu, whereas the blue solid curve represents the profile of the  $^{240}\text{Pu}$  line, which was recorded using a sample containing 9% Pu. The sharp peak on the rising edge of this curve presumably appeared owing to the transient Stark broadening and Doppler splitting effects reported in our previous study of Ce and U [31–32]. In this effect the signal of the neighboring abundant  $^{239}\text{Pu}$  overlaps with that of  $^{240}\text{Pu}$  owing to the presence of a high density of charged species and highly energetic species in the initial ablation plume. Fig. 9 shows the temporal variation in the isotopic abundance of  $^{240}\text{Pu}$  estimated from the profiles of the two lines of  $^{239}\text{Pu}$  and that of  $^{240}\text{Pu}$  recorded with a sample containing 27% Pu. The abundance of  $^{240}\text{Pu}$  was seemingly higher in the early stages and gradually decreased to reach a constant value of around 2.5% at a delay time of 10  $\mu$ s. This artificial variation arose from the abovementioned effects on the absorption spectrum. In some cases, however, a slight overlap of the  $^{239}\text{Pu}$  and  $^{240}\text{Pu}$  peaks remained even after these dynamic effects ceased. In the following Section (3.2.3), we will estimate the influence of such a residual overlap on the isotopic selectivity for three candidate transitions.

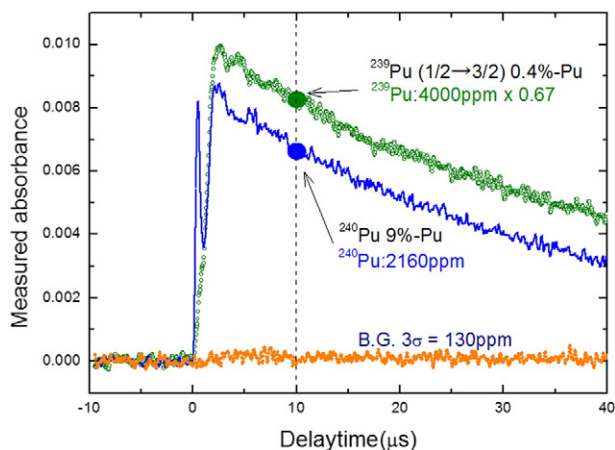
#### 3.2.3. Isotopic selectivity

If the separation between the  $^{240}\text{Pu}$  and  $^{239}\text{Pu}$  peaks in Figs. 5(a)–(c) is compared, it can be inferred that the transition at 420.6 nm has the lowest isotopic selectivity. To estimate the absorbance of  $^{239}\text{Pu}$  at the center frequency of the  $^{240}\text{Pu}$  peak, the two peaks of  $^{239}\text{Pu}$  in Fig. 5(a) were fitted with the multiple Voigt profile function (Eq. (6)), where the term number was limited to  $i = 2$  and Eq. (10) was used as a constraint condition. Fig. 10 shows the resulting fitted curve (red solid curve) together with the measured curve (blue filled circles). To enlarge the wings of each peak, the absorbance data are plotted on a logarithmic scale. From this figure, the absorbance of  $^{239}\text{Pu}$  at the center of the  $^{240}\text{Pu}$  peak was estimated to be 0.07. This means that 7.7% of the measured absorbance of  $^{240}\text{Pu}$  was due to the contribution of the  $^{239}\text{Pu}$  signal. It is

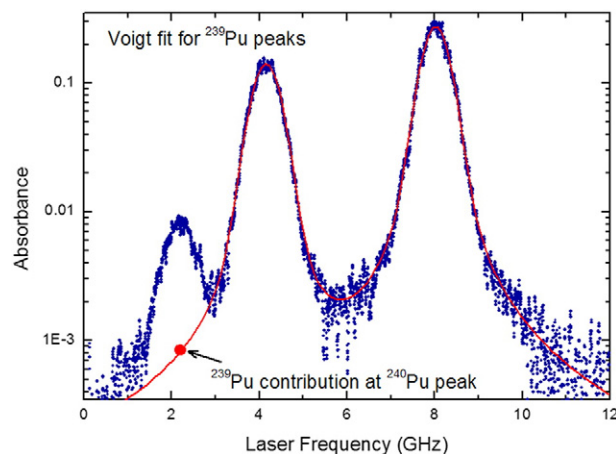
Table 4  
Isotope abundances determined from seven transitions based on Voigt profile fitting.

| $\lambda_{\text{air}}$<br>(nm) | $^{240}S$ | $^{239}S$ | Abundance (%) |      |
|--------------------------------|-----------|-----------|---------------|------|
|                                |           |           | 240           | 239  |
| 375.3628                       | 2.87      | 105.41    | 2.7           | 97.3 |
| 413.3005                       | 0.50      | 19.25     | 2.5           | 97.5 |
| 420.6481                       | 0.79      | 34.27     | 2.3           | 97.7 |
| 440.9510                       | 0.17      | 5.96      | 2.8           | 97.2 |
| 648.8853                       | 0.65      | 25.90     | 2.5           | 97.5 |
| 662.1215                       | 0.12      | 3.96      | 2.9           | 97.1 |
| 686.5482                       | 0.15      | 4.90      | 2.9           | 97.1 |
|                                |           | Mean      | 2.6           | 97.4 |
|                                |           | $1\sigma$ | 0.3           | 0.3  |





**Fig. 8.** Temporal absorption profiles for  $^{239}\text{Pu}$  ( $1/2 \rightarrow 3/2$ ) peak of 420.6-nm transition measured with a sample containing 0.4% Pu (green open circles) and that for  $^{240}\text{Pu}$  peak measured with a sample containing 9% Pu (blue solid curve) and that for background (orange filled circles). The background signal was measured by detuning the probe laser from the  $^{239}\text{Pu}$  ( $1/2 \rightarrow 3/2$ ) peak by approximately +4 GHz.



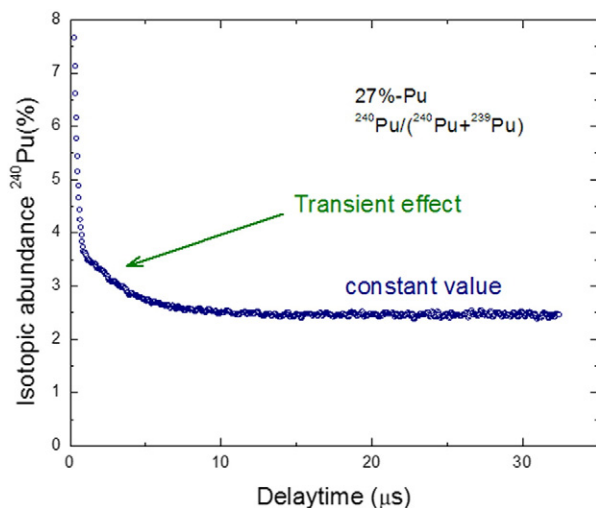
**Fig. 10.** Evaluation of contribution of absorption of  $^{239}\text{Pu}$  at resonant frequency of  $^{240}\text{Pu}$  for 420.6-nm transition. To enlarge the wing of each absorption peak, the vertical axis is taken on the logarithmic scale. The red solid curve represents the fitted spectrum of  $^{239}\text{Pu}$ , while the blue filled circles indicate the measured one (same as Fig. 3(a)).

noteworthy that this contribution fraction depended only on the isotopic ratio of Pu and was independent of the elemental concentration of Pu. Thus, a correction can be made by subtracting the contribution of  $^{239}\text{Pu}$  from the measured absorbance of  $^{240}\text{Pu}$ . As a result, the isotopic abundance of  $^{240}\text{Pu}$  was corrected to 2.3% for the Pu sample used in this study. It should be noted that this contribution has been eliminated in the method that uses the fitting parameters  $^{240}\text{S}$  and  $^{239}\text{S}$ . Hence, this corrected value is in agreement with that obtained by the former method.

From a similar analysis of the other candidate transitions, the contribution of  $^{239}\text{Pu}$  to the measured absorbance of the  $^{240}\text{Pu}$  peak was estimated to be several per cent for the transition at 375.4 nm and <0.5% for the transition at 648.9 nm. It should be noted that the precise determination of the isotopic abundance of Pu requires a comprehensive knowledge of its spectroscopic properties such as hfs and IS.

### 3.2.4. Signal-to-noise ratio and limit of detection

One of the major advantages of LAAS in comparison to LIBS is its high elemental and isotopic selectivity with negligible background, because



**Fig. 9.** Temporal variation in the isotopic abundance of  $^{240}\text{Pu}$  estimated from absorption profiles. Overestimation of  $^{240}\text{Pu}$  abundance in earlier stages arises from Stark broadening and Doppler splitting of absorption spectra.

of the use of cooled plasma. In Fig. 8, the orange filled circles labeled as BG denote the temporal absorption signal recorded with a sample containing 0.4% Pu by detuning the probe laser from the resonance of  $^{239}\text{Pu}$  ( $1/2 \rightarrow 3/2$ ) by approximately +4 GHz. A similar background signal was also obtained for the sample containing 9% Pu. From this figure, the signal-to-noise ratio was estimated to be approximately 60 for the  $^{240}\text{Pu}$  signal recorded with a sample containing 9% Pu (blue solid curve in Fig. 8). Furthermore, the LOD was determined from the standard deviation of the background ( $\sigma_{\text{BG}}$ ) and the conversion factor ( $K$ ) between absorbance and abundance on the basis of the  $3\sigma$  criterion:

$$\text{LOD} = \frac{3\sigma_{\text{BG}}}{K} \quad (12)$$

To avoid the transient effect, the absorbance at a delay time of 10  $\mu\text{s}$  in Fig. 8 was used for this estimation (blue and green filled circles). The blue curve represents the absorption signal of  $^{240}\text{Pu}$ , the isotopic abundance of which was 2.4%, as measured with the sample containing 9% Pu. Because the  $^{240}\text{Pu}$  concentration of the total weight of this sample was 0.216 wt%, the conversion factor  $K$  was calculated to be  $0.0067/2160 \text{ ppm}^{-1}$ . Therefore, the LOD for the detection of  $^{240}\text{Pu}$  in U-Pu MOX samples was estimated to be 120 ppm. A similar LOD was also obtained from the  $^{239}\text{Pu}$  ( $1/2 \rightarrow 3/2$ ) line. The concentrations of various Pu isotopes ( $^{239}\text{Pu}$ ,  $^{240}\text{Pu}$ ,  $^{241}\text{Pu}$ , and  $^{242}\text{Pu}$ ) in typical MOX fuels are expected to be higher than 0.1% [27–28], and thus this LOD matches the needs of analysis.

In general, the sensitivity of atomic absorption spectroscopy is proportional to the oscillator strength for each transition [49]. Thus, the transition at 375.4 nm, the absorbance of which is the strongest among the three candidate transitions, is expected to have the lowest LOD. Thus, we also determined the LOD of this transition. From the absorbance of the  $^{239}\text{Pu}$  ( $1/2 \rightarrow 3/2$ ) peak, the LOD was estimated to be 30 ppm using Eq. (12). This magnitude is reasonable, considering the strong absorbance of this transition (Table 3). In view of the inherently limited dynamic range of absorption measurements, it is of practical importance that the transition utilized is selected depending on the strength of absorbance.

### 3.2.5. Reproducibility and precision

For LA-based spectroscopic analysis, lower reproducibility is sometimes observed. To improve the reproducibility of LAAS, some researchers have proposed a method that measures the absorption signals of multiple isotopes simultaneously using multiple probe lasers



[13–15]. However, this method is not necessarily appropriate for remote analysis because of difficulty in directing the laser beams to an identical portion of the plume. For practical remote analysis, it is convenient that absorption signals are measured by repeatedly tuning the laser to the resonance of each isotope. Thus, we evaluated the reproducibility and precision of this method with the sample and transition used in Sections 3.2.2 and 3.2.3.

By storing the center frequencies of the three peaks in Fig. 5(a) using laser control software, the frequency of the probe laser was alternately switched among the three peaks. The temporal absorption profile with a delay time of up to 50  $\mu$ s was recorded with a digital oscilloscope with an integration time of 30 s (300 laser shots), and, from the magnitude of the absorbance at a delay time of 10  $\mu$ s ( $Abs(^{239}\text{Pu}, 1/2 \rightarrow 3/2)$ ,  $Abs(^{239}\text{Pu}, 1/2 \rightarrow 1/2)$ , and  $Abs(^{240}\text{Pu})$ ), the ratio  $r_i$  ( $i^{\text{th}}$  abundance of  $^{240}\text{Pu}$ ) was calculated according to Eq. (13), where  $Abs'(^{240}\text{Pu}) = Abs(^{240}\text{Pu}) \times 0.923$  (this correction factor is valid only for this specific pair of  $^{239}\text{Pu}$ – $^{240}\text{Pu}$  isotopic abundance) is a corrected absorbance value described in Section 3.2.4. Measurements were performed on three separate days, and the mean ( $R$ ), standard deviation ( $SD$ ), and RSD were calculated using Eqs. (14) and (15):

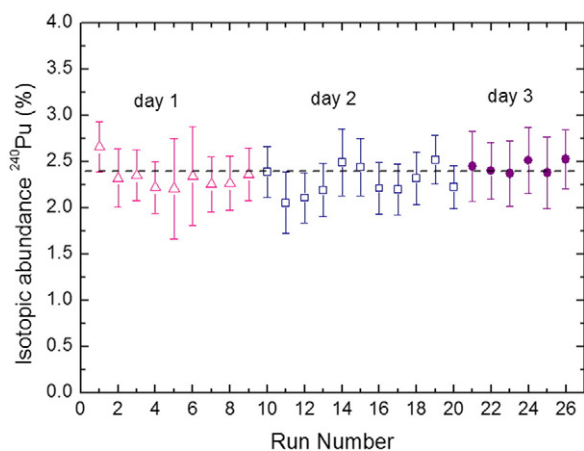
$$r_i = \frac{Abs'(^{240}\text{Pu})}{Abs(^{239}\text{Pu}, 1/2 \rightarrow 3/2) + Abs(^{239}\text{Pu}, 1/2 \rightarrow 1/2) + Abs'(^{240}\text{Pu})} \quad (13)$$

$$SD = \sqrt{\frac{\sum (R - r_i)^2}{n - 1}} \quad (14)$$

$$RSD = 100 \times \frac{SD}{R} \quad (15)$$

The results are summarized in Fig. 11. In this figure, the error bars represent uncertainties of  $2\sigma$  in the individual measurements, whereas the broken line represents the mean  $R$  of the measurement of  $2.34 \pm 0.15\%$ . Within the estimated range of error, this value is in good agreement with that determined from the fitting results described in Section 3.2.2.

From the results shown in Fig. 11, the RSD of the isotopic analysis of  $^{240}\text{Pu}$  was estimated to be 6%. The main sources of this variation are most likely to be shot-to-shot fluctuations in ablation pulse energy, temporal fluctuations in probe laser power, and a residual crater effect.



**Fig. 11.** Isotopic abundances of  $^{240}\text{Pu}$  measured for three different campaigns using 420.6-nm transition. Error bars represent  $2\sigma$  uncertainties and broken line denotes the weighted mean of  $^{240}\text{Pu}$  abundance. The number of replicate measurement is eight for each run.

## 4. Conclusions

We performed a spectroscopic study of Pu atoms and ions by employing the LAAS technique with a remote analysis system installed in a glove box. Absorption spectra were recorded for 17 transitions from the ground states of Pu atoms and ions to determine the absorbance, hyperfine constants, and IS. On the basis of this study, we selected three transitions of Pu atoms (375.4 nm, 420.6 nm, and 648.9 nm) as suitable for isotopic analysis by LAAS. Using these transitions, we investigated the analytical performance that was attainable, e.g., the linearity of the calibration curve, LOD, and RSD. As a result, a correlation coefficient  $R^2$  between absorbance and concentration of 0.9999, an LOD of 30–130 ppm, and an RSD of approximately 6% (for an abundance of  $^{240}\text{Pu}$  of 2.4%) were determined. Furthermore, it was confirmed that spectral interference and background arising from U atoms and ions were negligible in the spectra recorded with U–Pu MOX samples. These results indicate that LAAS is applicable to the remote isotopic analysis of highly radioactive nuclear fuels. To apply this analytical technique to actual radiotoxic fuels containing multiple actinide elements with dense and complicated spectroscopic structures, further spectroscopic studies on the other isotopes of Pu, Am and Cm will be needed.

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